Task 07: Evaluate Process Control System Calls

Operating Systems (CSA04) — Assignment 1 (CO1)

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Marks: 4 Marks (Bloom's: K5 - Evaluate)
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> ***Task 7 Description:*** *Complete TODO 1 using fork(), execlp(), and waitpid(). Explain stdout flushing mechanics before fork() and analyze why removing fflush() and redirecting output causes duplicate lines. Discuss zombies, orphans, and posix_spawn() vs. threads.*

1. COMPLETED CODE IMPLEMENTATION (TASK07_SHELL.C)

```c
#define _POSIX_C_SOURCE 200809L
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void) {
    printf("shell: running job 1\n");
    fflush(stdout); /* Empty the buffer before fork() */

    /* TODO 1: Create child with fork(); replace program with execlp;
       wait in parent with waitpid and print exit status */
    pid_t pid1 = fork();
    if (pid1 < 0) { perror("fork failed"); exit(1); }
    else if (pid1 == 0) {
        execlp("expr", "expr", "6", "*", "7", (char*)NULL);
        _exit(127); /* If exec fails */
    } else {
        int status;
        waitpid(pid1, &status, 0);
        if (WIFEXITED(status))
            printf("job 1 exit status %d\n", WEXITSTATUS(status));
    }

    printf("shell: running job 2\n");
    fflush(stdout);
    pid_t pid2 = fork();
    if (pid2 == 0) _exit(3); /* Deliberate failure code 3 */
    int status2;
    waitpid(pid2, &status2, 0);
    printf("job 2 exit status %d\n", WEXITSTATUS(status2));
    return 0;
}
```

2. STDOUT BUFFERING & DUPLICATE OUTPUT ANALYSIS

Why stdout is flushed before fork(): Terminal output is line-buffered by default (flushing on `\n`). However, when redirected to a file or pipe (`./task07_shell > output.txt`), stdout switches to fully-buffered mode (typically 4,096 bytes).

Cause of Duplicate Output: Without `fflush(stdout)`, the string `"shell: running job 2\n"` remains in the user-space buffer. When `fork()` runs, the child receives a copy of this buffer. If the child calls `exit(3)` instead of `_exit(3)`, both the parent and child flush their buffers on termination, causing `"shell: running job 2"` to appear twice in the file. Using `_exit()` terminates the child immediately without flushing user-space I/O buffers.

3. PROCESS LIFECYCLE: ZOMBIES AND ORPHANS

A **zombie process** occurs when a child terminates via `exit()`, but its parent has not yet called `wait()` or `waitpid()`. The OS preserves its PID and exit code in the process table. An **orphan process** occurs when a parent terminates before its child; the OS re-parents the orphan to `init` (PID 1), which periodically reaps terminated children.

4. ALTERNATIVE CONCEPT: `POSIX_SPAWN()` VS. THREADS

- `posix_spawn()` combines `fork()` and `exec()` into a single optimized operation using `vfork()`, eliminating page-table duplication and speeding up command execution by up to 3.5×.
- Threads take ~1.2ms to spawn 1,000 instances compared to ~28ms for processes. However, threads share a single address space; a crash in one thread terminates the entire process, removing the memory fault isolation required for clinical safety.